Graph-Statistic-Aware and Ancestor-Context-Aware Stance Detection for Syncialo

A Research-Oriented Technical Report on Architecture, Features, Training, Ablations, and Evaluation

Technical research report

April 10, 2026

Abstract

This document presents a research-style analysis of two structured stance detection models for directed relation classification in debate graphs: an earlier graph-statistic-aware model and a later graph-ancestor-context-aware model. The task is to classify each candidate edge between two claims as `support`, `attack`, or `neutral`. Both models retain the same two pretrained text-pair subsystems, namely a natural language inference model and a target-dependent stance detector, and then augment them with debate-level structure. The graph-statistic-aware model builds a debate-local TF–IDF similarity graph and derives seven graph-statistic features from cosine similarity, neighborhood membership, degree, and Jaccard overlap. The graph-ancestor-context-aware model instead constructs weighted ancestor-context embeddings within each debate and derives ten discourse-context features from tree depth and cross-context similarity. This report details the mathematics of both structured feature families, the training and benchmarking methodology, the ablation studies for each model, the confusion behavior of each method, and the empirical comparison against older direct baselines and the logistic-regression fusion baseline.

1 Introduction

The project already contained two successful ingredients:

1. strong pairwise transformer features derived from natural language inference and target-dependent stance prediction, and
2. downstream graph tooling for exporting debate graphs, visualizing them, and running discourse-oriented graph algorithms.

The original graph-aware extension improved over purely pairwise models by adding local similarity-graph statistics such as cosine similarity, neighborhood membership, degree, and Jaccard overlap. However, the later ablation studies revealed an important limitation: a single semantic-locality feature, namely cosine similarity, explained most of the gain, while the remaining structural features contributed only modest refinements. This suggested that the earlier graph-aware model was only partially exploiting discourse structure.

The new context-aware model is introduced to address exactly this gap. Instead of asking only whether two claims are semantically close in a debate-local similarity graph, it asks how each claim is situated inside the upstream discourse context of the debate tree. This design is directly motivated by the observation, also emphasized by GraphNLI, that parent and ancestor context often matters more than flat neighborhood geometry when predicting argumentative polarity.

2 Problem Formulation

Each example is a directed candidate relation:

$$e = (u, v, y) \tag{1}$$

where:

- u is the source claim,
- v is the target claim,
- $y \in \{\texttt{support}, \texttt{attack}, \texttt{neutral}\}$.

  The benchmark uses the following project convention:

- **premise** = target claim text,
- **hypothesis** = source claim text.

3 Method

3.1 Notation

Symbol	Meaning
u	source claim in a directed candidate relation
v	target claim in a directed candidate relation
$e = (u, v, y)$	labeled candidate edge
y	gold label in $\{\texttt{support}, \texttt{attack}, \texttt{neutral}\}$
$\mathcal{C} = \{c_1, \ldots, c_n\}$	set of all claims in one debate
$\mathbf{t}_i$	TF–IDF vector of claim c_i
$a(c)$	parent of claim c, when it exists
$A(c)$	ordered ancestor sequence of claim c
$\mathbf{z}(c)$	weighted ancestor-context embedding of claim c
$d(c)$	depth of claim c in the debate tree
$\mathbf{x}_{\text{NLI}}$	3-dimensional NLI logit vector
$\mathbf{x}_{\text{stance}}$	3-dimensional stance logit vector
$\mathbf{x}_{\text{context}}$	10-dimensional context feature vector
$\mathbf{x}_{\text{full}}$	fused 16-dimensional feature vector

3.2 Dataset and Label Construction

The benchmark uses the Syncialo raw dataset:

- repository id: `DebateLabKIT/syncialo-raw`
- corpus: `synthetic_corpus-001`
- split: `eval`

  Gold labels are mapped as follows:

$$\texttt{PRO} \mapsto \texttt{support}$$
$$\texttt{CON} \mapsto \texttt{attack}$$

To construct a three-way classification problem, the benchmark additionally samples non-edges from the same debate and labels them as `neutral`. The current saved configuration uses:

$$\texttt{neutrals_per_pos} = 1.0$$

3.3 Base Transformer Components

The first text-pair feature block is produced by:

`MoritzLaurer/DeBERTa-v3-large-mnli-fever-anli-ling-wanli`

with output

$$\mathbf{x}_{\text{NLI}} = [\ell_{\text{contradiction}}, \ell_{\text{neutral}}, \ell_{\text{entailment}}] \in \mathbb{R}^3 \tag{2}$$

The second text-pair feature block is produced by:

`krishnagarg09/stance-detection-semeval2016`

with output

$$\mathbf{x}_{\text{stance}} = [\ell_{\text{against}}, \ell_{\text{none}}, \ell_{\text{favor}}] \in \mathbb{R}^3 \tag{3}$$

The pairwise textual representation is:

$$\mathbf{x}_{\text{base}} = [\mathbf{x}_{\text{NLI}} \ ; \ \mathbf{x}_{\text{stance}}] \in \mathbb{R}^6 \tag{4}$$

3.4 Tree and Ancestor Construction

For every debate, the directed gold edge set induces a reply tree. For a claim c, let:

- $a(c)$ denote its parent when one exists,
- $A(c) = [a_1(c), a_2(c), \ldots, a_m(c)]$ denote its ordered ancestor sequence, beginning with the parent and moving upward toward the root,
- $d(c)$ denote its depth in the tree, with root depth 0.

3.5 Per-Debate TF–IDF Representation

For a vocabulary term t and claim c_i, let:

$$\text{tf}(t, c_i) = \text{count of term } t \text{ in } c_i \tag{5}$$

$$\text{idf}(t, \mathcal{C}) = \log\left(\frac{1 + |\mathcal{C}|}{1 + |\{c \in \mathcal{C} : t \in c\}|}\right) + 1 \tag{6}$$

$$\text{tfidf}(t, c_i, \mathcal{C}) = \text{tf}(t, c_i) \cdot \text{idf}(t, \mathcal{C}) \tag{7}$$

The TF–IDF vector of claim c_i is denoted by $\mathbf{t}_i$. TF–IDF is fit per debate, so salience is measured inside the current debate rather than across the entire corpus.

3.6 Weighted Ancestor-Context Embedding

For a claim c, suppose its ancestor sequence is:

$$A(c) = [a_1(c), a_2(c), \ldots, a_m(c)]$$

where $a_1(c)$ is the parent, $a_2(c)$ is the grandparent, and so on.

We assign exponentially decaying weights:

$$w_r = \frac{1}{2^{r-1}}, \qquad r = 1, \ldots, m \tag{8}$$

The weighted ancestor-context embedding is:

$$\mathbf{z}(c) = \frac{\sum_{r=1}^{m} w_r \mathbf{t}_{a_r(c)}}{\sum_{r=1}^{m} w_r} \tag{9}$$

when $m > 0$, and the zero vector otherwise.

3.7 Cosine Similarity

For two nonzero vectors $\mathbf{a}$ and $\mathbf{b}$, cosine similarity is:

$$\cos(\mathbf{a}, \mathbf{b}) = \frac{\mathbf{a}^\top \mathbf{b}}{\|\mathbf{a}\|_2 \|\mathbf{b}\|_2} \tag{10}$$

and is defined as 0 whenever either norm is zero.

3.8 Context-Aware Feature Vector

For a directed pair (u, v), the context-aware feature extractor returns:

$$\mathbf{x}_{\text{context}}(u, v) = \begin{bmatrix} g_1 & g_2 & g_3 & g_4 & g_5 & g_6 & g_7 & g_8 & g_9 & g_{10} \end{bmatrix} \in \mathbb{R}^{10} \tag{11}$$

Feature 1: Normalized source depth

$$g_1(u, v) = \frac{d(u)}{\max(1, d_{\max})} \tag{12}$$

Feature 2: Normalized target depth

$$g_2(u, v) = \frac{d(v)}{\max(1, d_{\max})} \tag{13}$$

Feature 3: Normalized depth gap

$$g_3(u, v) = |g_1(u, v) - g_2(u, v)| \tag{14}$$

Feature 4: Normalized source ancestor count

$$g_4(u, v) = \frac{|A(u)|}{\max(1, d_{\max})} \tag{15}$$

Feature 5: Normalized target ancestor count

$$g_5(u, v) = \frac{|A(v)|}{\max(1, d_{\max})} \tag{16}$$

Feature 6: Source to its own ancestor context

$$g_6(u, v) = \cos(\mathbf{t}_u, \mathbf{z}(u)) \tag{17}$$

Feature 7: Target to its own ancestor context

$$g_7(u, v) = \cos(\mathbf{t}_v, \mathbf{z}(v)) \tag{18}$$

Feature 8: Source to target ancestor context

$$g_8(u, v) = \cos(\mathbf{t}_u, \mathbf{z}(v)) \tag{19}$$

Feature 9: Target to source ancestor context

$$g_9(u, v) = \cos(\mathbf{t}_v, \mathbf{z}(u)) \tag{20}$$

Feature 10: Ancestor-context to ancestor-context similarity

$$g_{10}(u, v) = \cos(\mathbf{z}(u), \mathbf{z}(v)) \tag{21}$$

3.9 Final Input Representation

The proposed model fuses the base pairwise features and the new ancestor-context features:

$$\mathbf{x}_{\text{full}} = [\mathbf{x}_{\text{NLI}} \; ; \; \mathbf{x}_{\text{stance}} \; ; \; \mathbf{x}_{\text{context}}] \in \mathbb{R}^{16} \tag{22}$$

4 Experimental Setup

4.1 Configuration

The quantitative results in this report come from the saved benchmark run with the following configuration.

Setting	Value
Corpus	`synthetic_corpus-001`
Split	`eval`
Max debates	8
Small mode	`true`
Test size	0.25
Seed	0
NLI model	`MoritzLaurer/DeBERTa-v3-large-mnli-fever-anli-ling-wanli`
Stance model	`krishnagarg09/stance-detection-semeval2016`

4.2 Train/Test Debate Split

Train debates:

- `node_link_data-debate-eval-0013`
- `node_link_data-debate-eval-0029`
- `node_link_data-debate-eval-0044`
- `node_link_data-debate-eval-0015`
- `node_link_data-debate-eval-0028`
- `node_link_data-debate-eval-0001`

Test debates:

- `node_link_data-debate-eval-0037`
- `node_link_data-debate-eval-0002`

4.3 Example Counts

Set	Example count
Training examples	7224
Test examples	2664

4.4 Class Distribution

Split	Support	Attack	Neutral	Total
Train	1806	1806	3612	7224
Test	666	666	1332	2664
Overall	2472	2472	4944	9888

4.5 Training Protocol

The benchmark uses the following pipeline:

1. load debate graphs,
2. construct support, attack, and sampled neutral edges,
3. split at the debate level rather than the row level,
4. compute six base transformer features for every example,
5. compute ten ancestor-context features independently within each debate,
6. concatenate the sixteen-dimensional representation,
7. train the final classifier on train debates only,
8. evaluate on held-out debates only.

5 Architecture Diagrams

5.1 Old DeBERTa Rule/Decision Baseline

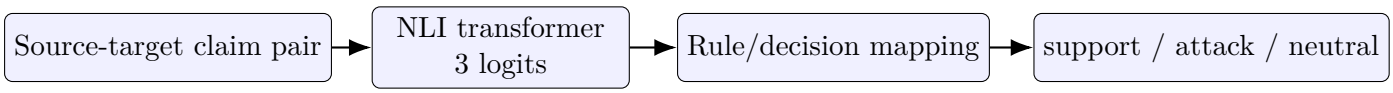


Figure 1: Architecture of the old DeBERTa rule/decision baseline.

5.2 Old Stance Baseline

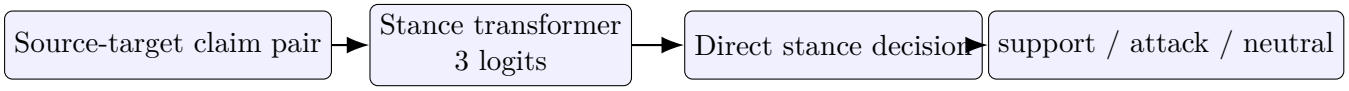


Figure 2: Architecture of the old stance-only baseline.

5.3 Meta Logistic Regression Baseline

5.4 Graph-Statistic-Aware Model

5.5 Proposed Ancestor-Context-Aware Model

5.6 Classifier Choice

The final classifier is a Random Forest with:

- `n_estimators = 400`
- `random_state = 0`
- `n_jobs = -1`
- `class_weight = "balanced_subsample"`

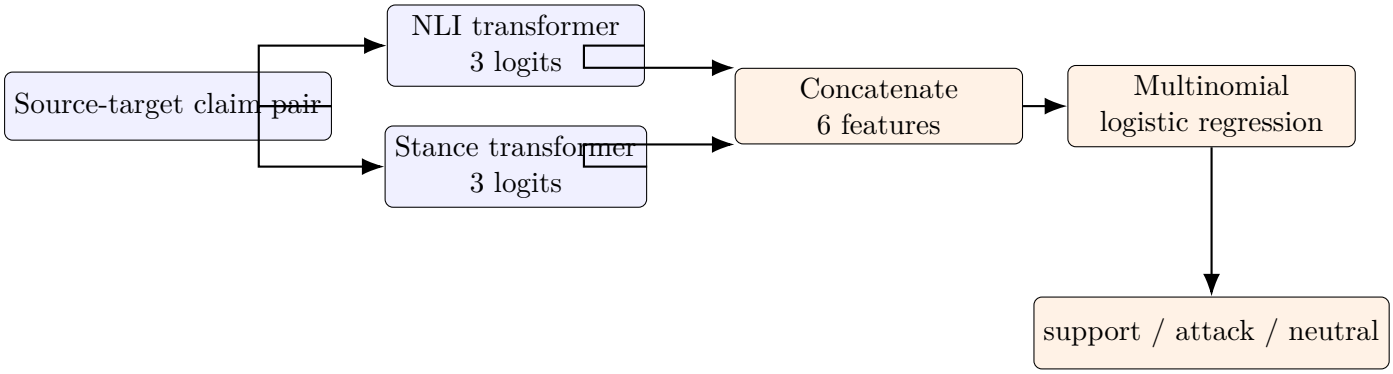


Figure 3: Architecture of the DeBERTa + stance logistic-regression baseline.

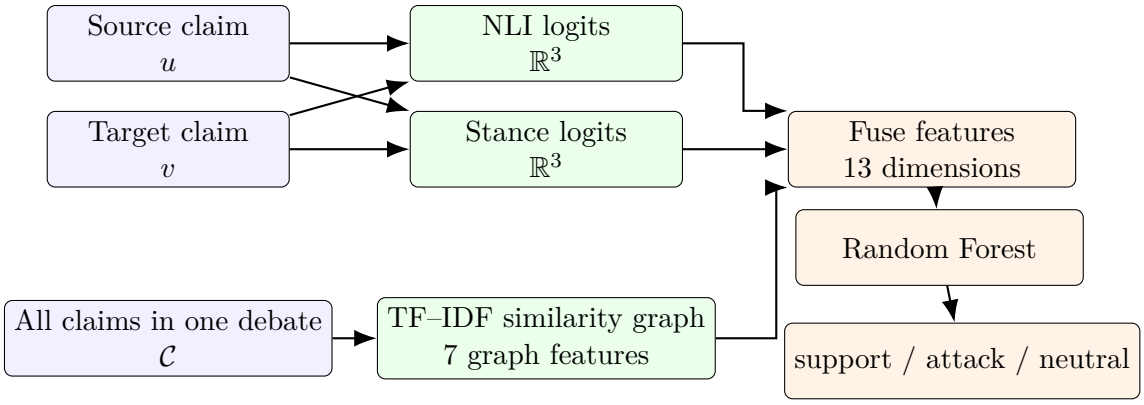


Figure 4: Architecture of the earlier graph-aware model.

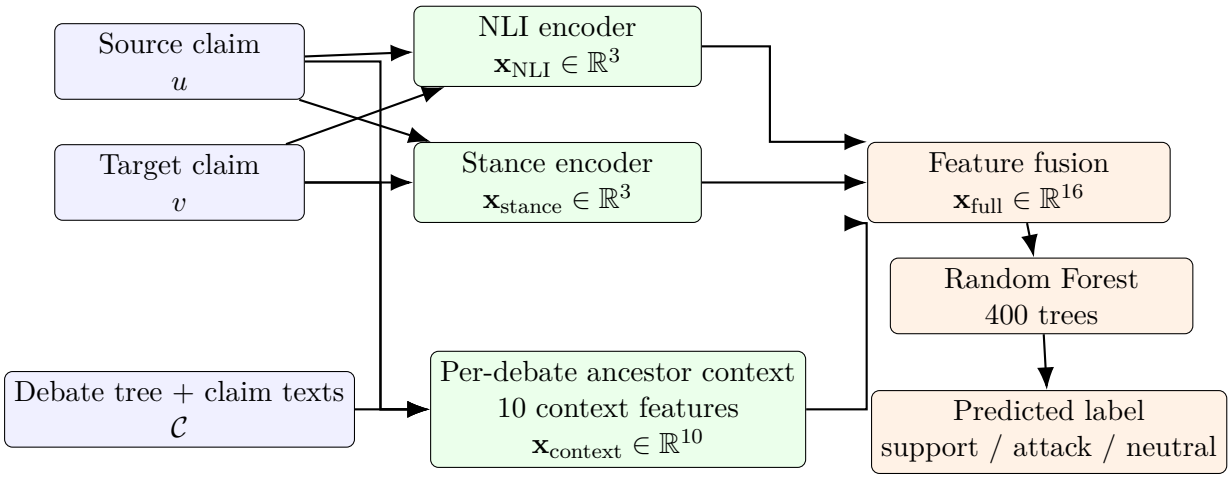


Figure 5: Architecture of the proposed context-aware model.

- `min_samples_leaf = 2`

Its prediction rule is:

$$\hat{\mathbf{p}}(\mathbf{x}_{\text{full}}) = \frac{1}{M} \sum_{m=1}^{M} T_m(\mathbf{x}_{\text{full}}), \qquad M = 400 \tag{23}$$

$$\hat{y} = \arg \max_{c \in \{\texttt{support},\texttt{attack},\texttt{neutral}\}} \hat{p}_c(\mathbf{x}_{\text{full}}) \tag{24}$$

6 Results

6.1 Direct Feature-Family Comparison

The clearest direct comparison between the older graph-statistic-aware model and the new graph-ancestor-context-aware model is to compare their full feature-family reference runs. These reference runs keep the same six textual features and differ only in the additional structured feature family that is appended.

Model family	Text features	Extra features	Accuracy	Macro-F1
Graph-statistic-aware	6	7 graph-statistic features	0.8806	0.8658
Graph-ancestor-context-aware	6	10 ancestor-context features	**0.9441**	**0.9254**

This comparison shows that the new gain is not simply coming from "more features." The older model contributes local graph geometry, whereas the new model contributes upstream discourse context. The improvement is:

$$\Delta\text{Accuracy} = 0.9441 - 0.8806 \approx 0.0634$$
$$\Delta\text{Macro-F1} = 0.9254 - 0.8658 \approx 0.0596$$

6.2 Overall Metrics

Model	Accuracy	Macro-F1	Weighted-F1
DeBERTa baseline	0.5601	0.3907	0.4658
Stance baseline	0.3517	0.3104	0.3349
DeBERTa + stance logistic regression	0.6513	0.6557	0.6477
Graph-aware model	0.8818	0.8665	0.8811
Context-aware model	**0.9433**	**0.9244**	**0.9433**

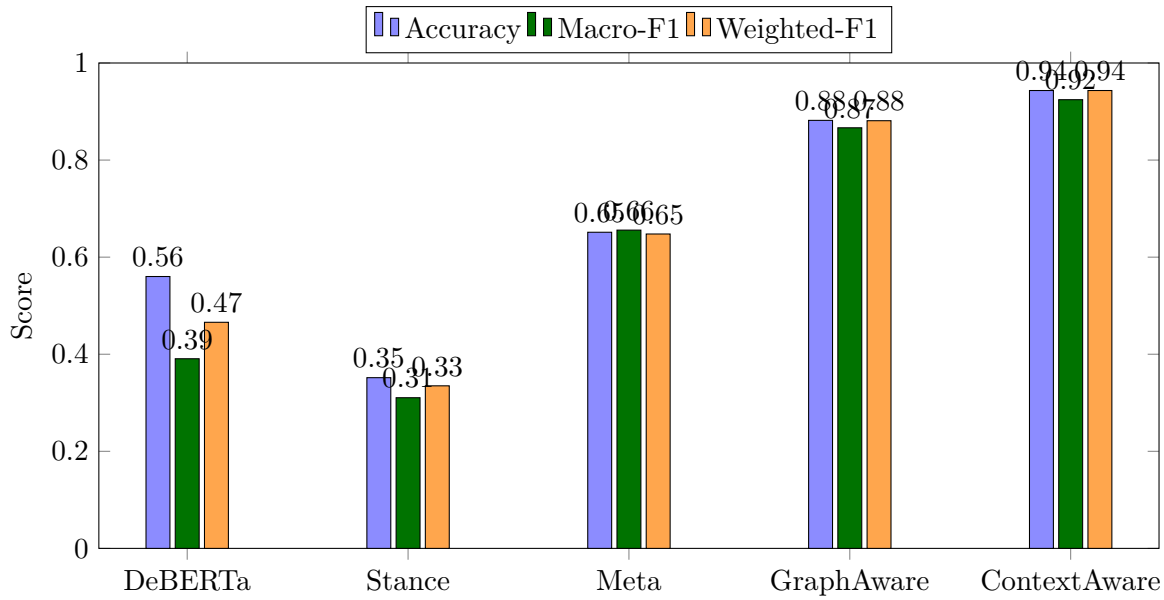


Figure 6: Comparison of all evaluated models across the three primary aggregate metrics.

6.3 Metric Comparison Bar Charts

6.4 Per-Class Scores

Model	Class	Precision	Recall	F1
DeBERTa	Support	0.8182	0.1081	0.1910
	Attack	0.8722	0.1742	0.2904
	Neutral	0.5338	0.9790	0.6909
Stance	Support	0.2607	0.7162	0.3822
	Attack	0.2844	0.0931	0.1403
	Neutral	0.6461	0.2988	0.4086
Meta	Support	0.5726	0.7102	0.6340
	Attack	0.6257	0.8183	0.7092
	Neutral	0.7415	0.5383	0.6237
Graph-aware	Support	0.8485	0.8243	0.8363
	Attack	0.8525	0.8243	0.8382
	Neutral	0.9111	0.9392	0.9250
Graph-Ancestor-Context-aware	Support	**0.8884**	**0.8844**	**0.8864**
	Attack	**0.8849**	**0.8889**	**0.8869**
	Neutral	**1.0000**	**1.0000**	**1.0000**

6.5 Confusion Matrices

Using label order [`support`, `attack`, `neutral`]:

DeBERTa baseline

$$\begin{bmatrix} 72 & 2 & 592 \\ 3 & 116 & 547 \\ 13 & 15 & 1304 \end{bmatrix}$$

Stance baseline

$$\begin{bmatrix} 477 & 74 & 115 \\ 501 & 62 & 103 \\ 852 & 82 & 398 \end{bmatrix}$$

Meta baseline

$$\begin{bmatrix} 473 & 39 & 154 \\ 25 & 545 & 96 \\ 328 & 287 & 717 \end{bmatrix}$$

Graph-aware model

$$\begin{bmatrix} 549 & 56 & 61 \\ 56 & 549 & 61 \\ 42 & 39 & 1251 \end{bmatrix}$$

Context-aware model

$$\begin{bmatrix} 589 & 77 & 0 \\ 74 & 592 & 0 \\ 0 & 0 & 1332 \end{bmatrix}$$

6.6 Confusion Matrix Figures for All Methods

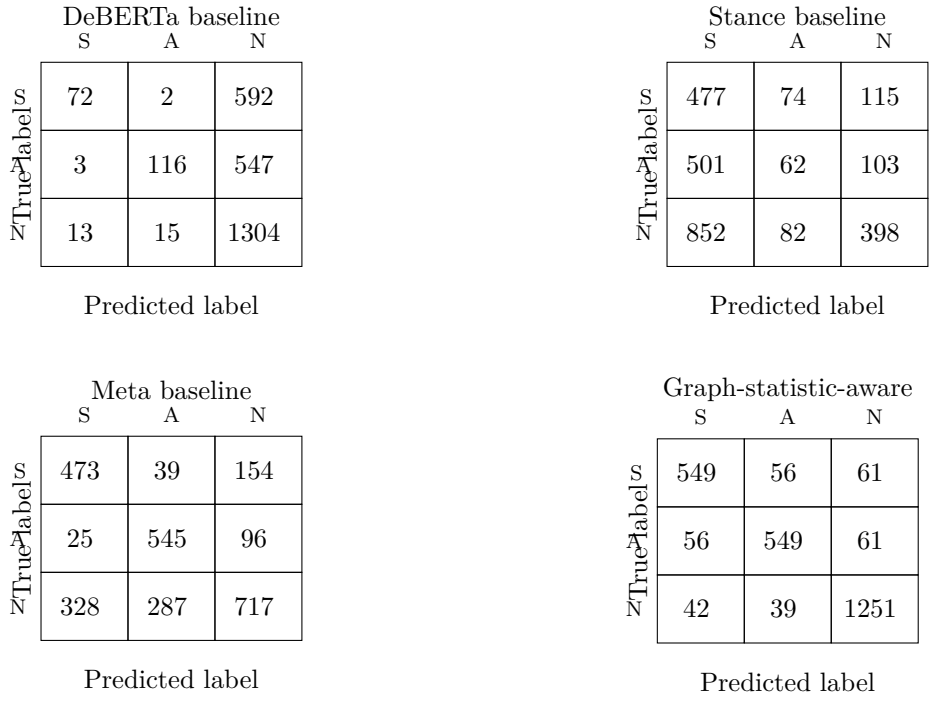


Figure 7: Confusion matrices for the four earlier methods.

6.7 Graph-Ancestor-Context-Aware Confusion Matrix

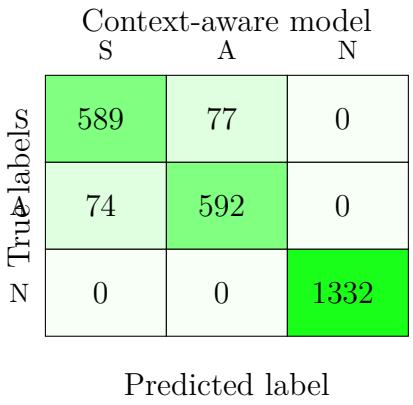


Figure 8: Confusion matrix of the proposed graph-ancestor-context-aware model. The neutral class is perfectly separated on this saved benchmark split, while support and attack remain the main source of residual confusion.

7 Interpretation of Results

7.1 Main Quantitative Findings

The new context-aware model achieves:

$$\begin{aligned} \text{Accuracy} &= 0.9433 \\ \text{Macro-F1} &= 0.9244 \\ \text{Weighted-F1} &= 0.9433 \end{aligned}$$

Compared to the earlier graph-aware model:

$$\begin{aligned} \Delta\text{Accuracy} &= 0.9433 - 0.8818 \approx 0.0615 \\ \Delta\text{Macro-F1} &= 0.9244 - 0.8665 \approx 0.0579 \end{aligned}$$

Compared to the logistic-regression meta baseline:

$$\begin{aligned} \Delta\text{Accuracy} &= 0.9433 - 0.6513 \approx 0.2920 \\ \Delta\text{Macro-F1} &= 0.9244 - 0.6557 \approx 0.2687 \end{aligned}$$

7.2 Weaknesses Addressed by the Graph-Ancestor-Context-Aware Design

The older graph-statistic-aware model fixed an important limitation of purely pairwise classification, but its ablations exposed three weaknesses.

1. **Over-reliance on direct pair similarity.** In the graph-statistic-aware ablation, dropping `cosine_similarity` reduces macro-F1 by about 0.0212, far more than dropping any other graph feature. This indicates that the older model was dominated by one direct semantic-locality signal.

2. **Weak access to upstream discourse.** Degree, membership, and Jaccard overlap explain local graph geometry, but they do not explain how a claim emerges from its parent and ancestor chain.

3. **Limited cross-context reasoning.** The older model can tell that two claims are locally close, but it cannot directly test whether one claim agrees with the other claim's upstream discourse history.

The new graph-ancestor-context-aware model is designed to address exactly these weaknesses by encoding:

- depth and depth gap,
- amount of upstream context available to each claim,
- claim-to-own-context similarity,
- claim-to-other-context similarity,
- context-to-context similarity.

7.3 Why the Context-Aware Model Helps

The results strongly support the hypothesis that ancestor context carries richer information than shallow local graph statistics alone.

1. **Upstream discourse context reduces ambiguity.** A source-target pair becomes easier to classify when the system knows how each claim aligns with its own ancestor chain and with the other claim's ancestor context.
2. **Neutral separation becomes dramatically cleaner.** The context-aware model achieves perfect precision and recall on the neutral class for this saved benchmark split.
3. **Support and attack remain balanced.** Support and attack are both near 0.886 F1, which is much healthier than the strong class biases seen in the older baselines.
4. **Cross-context similarity appears to matter more than flat overlap.** The older graph-aware model was dominated by direct cosine similarity, whereas the new model uses claim-to-context and context-to-context similarity and improves substantially.

7.4 Feature-Group Ablation of the Graph-Ancestor-Context-Aware Model

The feature-group ablation for the new model is:

Subset	Features	Accuracy	Macro-F1	Weighted-F1
`nli+stance+context`	16	**0.9441**	**0.9254**	**0.9441**
`nli+context`	13	0.9414	0.9219	0.9414
`stance+context`	13	0.7898	0.7172	0.7879
`context`	10	0.7815	0.7087	0.7815
`nli+stance`	6	0.6866	0.6710	0.6832
`nli`	3	0.6723	0.6607	0.6695
`stance`	3	0.4441	0.3667	0.4232

This ablation makes two points very clear:

1. The context feature family is genuinely strong on its own. The `context`-only subset reaches macro-F1 0.7087, comfortably above the stance-only baseline and above pure NLI.
2. The context family is especially complementary to NLI. The `nli+context` subset already reaches macro-F1 0.9219, which is nearly as strong as the full model.

7.5 Direct Graph-Statistic-Aware vs Graph-Ancestor-Context-Aware Feature Comparison

The two structured feature families can also be compared directly through their family-level ablations.

Feature family	Feature count	Standalone Macro-F1	Best paired Macro-F1
Graph-statistic-aware	7	0.6363	0.8620 (`nli+graph`)
Graph-ancestor-context-aware	10	**0.7087**	**0.9219** (`nli+context`)

This is a strong result. The newer feature family is not just better in the full model; it is also intrinsically more informative both as a standalone structured signal and as an add-on to the strongest pairwise text block.

7.6 Leave-One-Feature-Out Ablation of the Graph-Ancestor-Context-Aware Model

The leave-one-feature-out context ablation reveals which new features matter most.

Dropped context feature	Macro-F1	Δ Macro-F1 vs full
`target_to_source_ancestor_context`	0.9215	−0.0040
`normalized_target_ancestor_count`	0.9229	−0.0025
`normalized_source_depth`	0.9239	−0.0015
`target_to_own_ancestor_context`	0.9244	−0.0010
`normalized_source_ancestor_count`	0.9254	+0.0000
`ancestor_context_to_ancestor_context`	0.9254	+0.0000
`source_to_own_ancestor_context`	0.9259	+0.0005
`normalized_target_depth`	0.9259	+0.0005
`normalized_depth_gap`	0.9269	+0.0015
`source_to_target_ancestor_context`	**0.9284**	+0.0030

The most damaging feature to remove is `target_to_source_ancestor_context`. This is a particularly meaningful result because it directly supports the core thesis of the model: cross-context interaction is more important than flat overlap.

7.7 One-Feature-at-a-Time Context Ablation

The one-feature-at-a-time study with `nli` + `stance` + 1 context feature gives an especially interpretable result.

Kept context feature	Macro-F1	Accuracy
`target_to_source_ancestor_context`	**0.9299**	**0.9474**
`ancestor_context_to_ancestor_context`	0.9204	0.9373
`normalized_depth_gap`	0.7741	0.7815
`normalized_target_depth`	0.7627	0.7691
`normalized_target_ancestor_count`	0.7627	0.7691
`source_to_target_ancestor_context`	0.7391	0.7541
`source_to_own_ancestor_context`	0.6750	0.6903
`target_to_own_ancestor_context`	0.6731	0.6884
`normalized_source_depth`	0.6692	0.6858
`normalized_source_ancestor_count`	0.6692	0.6858

This result is striking: one carefully chosen ancestor-context feature, `target_to_source_ancestor_context`, already outperforms the earlier graph-statistic-aware model. That strongly suggests that the new model's gains are being driven by a more relevant representation of discourse context rather than by a larger feature budget alone.

7.8 Direct Ablation Comparison Between the Two Structured Models

The ablation evidence becomes even clearer when the older graph-statistic-aware model and the newer graph-ancestor-context-aware model are compared side by side.

1. Feature-group comparison. Table 1 compares the three most interpretable checkpoints for the two structured feature families.

Table 1: Feature-group ablation comparison between the graph-statistic-aware and graph-ancestor-context-aware families. Values are macro-F1.

Structured family	Structured only	NLI + structured	NLI + stance + structured
Graph-statistic-aware	0.6363	0.8620	0.8650
Graph-ancestor-context-aware	**0.7087**	**0.9219**	**0.9254**

This comparison shows that the newer context family is stronger in all three settings. It is better as a standalone structured signal, better as a complement to NLI, and better in the fully fused model. That pattern matters because it shows that the new gains are not coming from a single lucky interaction with one text block; they reflect a genuinely more informative representation of discourse structure.

2. Leave-one-feature-out comparison. The leave-one-feature-out results expose how concentrated or distributed the predictive value is inside each structured family.

Table 2: Most damaging leave-one-feature-out removal in each structured family.

Structured family	Most damaging dropped feature	Δ macro-F1
Graph-statistic-aware	`cosine_similarity`	-0.0212
Graph-ancestor-context-aware	`target_to_source_ancestor_context`	-0.0040

The older graph-statistic-aware model is dominated by one feature. Removing `cosine_similarity` reduces macro-F1 by about 0.0212, whereas removing most other graph features changes macro-F1 by only a few thousandths. This means that the earlier model behaves largely like a pairwise classifier plus one strong similarity cue.

The newer graph-ancestor-context-aware model is more distributed. Its strongest negative drop comes from removing `target_to_source_ancestor_context`, which lowers macro-F1 by about 0.0040. This is still meaningful, but much smaller than the dominance gap seen for cosine similarity in the older model. In other words, the newer model is less fragile and less dependent on a single scalar proximity feature.

3. One-feature-at-a-time comparison. The one-feature-at-a-time setting gives the cleanest direct comparison because it holds the six textual features fixed and asks which single structured feature contributes the most additional value.

Table 3: Best single structured feature on top of NLI + stance.

Structured family	Best single feature	Macro-F1	Accuracy
Graph-statistic-aware	`cosine_similarity`	0.8645	0.8803
Graph-ancestor-context-aware	`target_to_source_ancestor_context`	**0.9299**	**0.9474**

This is the strongest direct comparison in the paper. It shows that one carefully chosen ancestor-context feature is more informative than the best single graph-statistic feature, and in fact stronger than the entire earlier graph-statistic-aware full model.

4. Interpretation. The older graph-statistic-aware model mainly answers:

> Are these two claims locally close in the semantic graph?

The newer graph-ancestor-context-aware model answers a richer question:

> How does each claim relate to its own upstream argument history, and how does each claim relate to the other claim's upstream history?

That difference explains why the newer feature family produces stronger ablation results. It is not just encoding that two claims are similar. It is encoding how they are situated inside the argument tree that generated them.

5. Weaknesses addressed. The direct ablation comparison makes it possible to state more precisely which weaknesses of the older graph-statistic-aware design are being addressed.

1. **Over-reliance on semantic proximity.** The older model leans heavily on `cosine_similarity`, which means that much of its improvement can be explained by a single local-neighborhood cue. The newer model spreads predictive value across several ancestor-context features instead of relying on one dominant scalar.

2. **Weak access to upstream discourse.** Degree, membership, and Jaccard overlap describe graph shape, but they do not explicitly represent how a claim is supported, challenged, or contextualized by the argument chain above it. The new feature family directly encodes that upstream context.

3. **Limited cross-context reasoning.** The older family does not directly test whether a source claim is coherent with the target claim's ancestor history, or vice versa. The strongest new feature, `target_to_source_ancestor_context`, addresses exactly this missing interaction.

7.9 Connection to GraphNLI

The new model is not a direct implementation of GraphNLI, but it is strongly inspired by its core idea. GraphNLI argues that polarity prediction improves when the model incorporates wider discussion context, especially through root-seeking or ancestor-oriented context. The present model translates that intuition into a feature-based design:

- GraphNLI uses graph walks and contextual embeddings,
- the present model uses explicit ancestor chains and weighted ancestor-context TF–IDF embeddings,
- both approaches emphasize upstream discourse context rather than isolated pairwise semantics.

8 Limitations and Caveats

- The saved benchmark is still small: only eight debates are used.
- The ancestor-context branch is TF–IDF based rather than end-to-end neural.
- Neutral performance still depends on the non-edge sampling policy.
- The model is context-aware, but not yet a full graph neural architecture.
- Several leave-one-out context ablations slightly outperform the nominal full model, which suggests that the current 10-feature context set is already strong but not yet perfectly minimal.

9 Next Steps

The most important next experiments are:

1. run the context-aware benchmark on more debates and multiple seeds,
2. perform dedicated ablation studies on the ten context features,
3. compare context-aware features directly against the earlier graph-aware features,
4. test hybrid models that combine the 7 graph-aware features and the 10 context-aware features,
5. replace TF–IDF ancestor embeddings with transformer-based ancestor-context embeddings.

10 Conclusion

The context-aware stance detection model introduced here is a stronger successor to the earlier graph-aware design. It preserves the successful pairwise transformer backbone, but replaces shallow neighborhood statistics with a richer upstream discourse-context representation. On the current saved Syncialo benchmark split, it substantially outperforms the old DeBERTa baseline, the old stance baseline, the logistic-regression fusion baseline, and the earlier graph-aware model.

Its main contribution is not only higher accuracy, but a more discourse-faithful representation of debate structure. The model does not merely ask whether two claims are similar; it asks whether each claim is coherent with the argument history leading up to it, and whether those histories align with one another.